

Module-End -2

Module End Assignment:

“Data Transformation, Modeling & Visualization with Power BI”

Assignment Task Overview

Title: Analyzing Smart City Bike Sharing Data using Power BI

Problem Statement:

The objective of this project is to analyze bike-sharing system performance by examining revenue, distance usage, rental cost, and station utilization across different cities, stations, and years. The analysis aims to identify high-performing and underperforming stations, understand cost efficiency, and evaluate the impact of operational factors such as active status, banking, and bonus availability to support better operational and business decision-making.

Dataset:

Import and explore the dataset in Power BI.

Clean and transform data using Power Query.

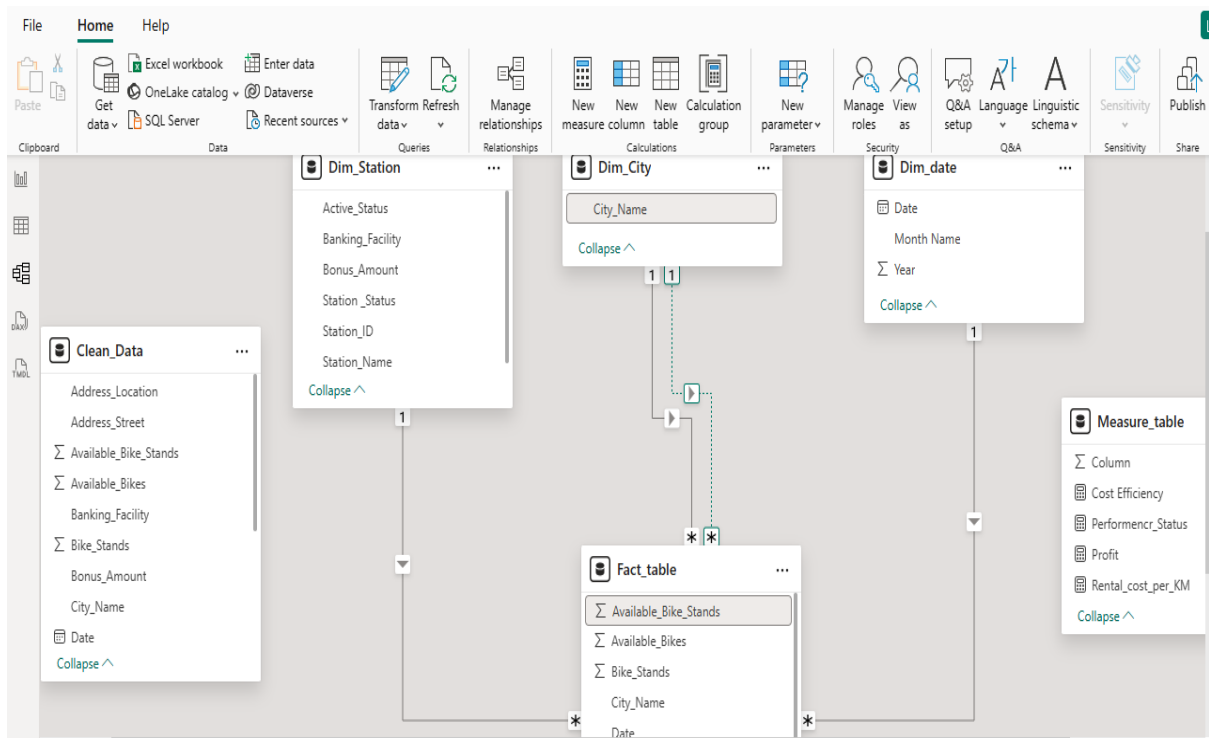
The screenshot displays the Power Query Editor interface. The main area shows a table with 23 rows and 4 columns: Station_ID, Station_Name, Address_Street, and Address_Location. The table contains data for various stations, including 'Test Dsi Plaisir', 'St Férréol Davso', 'Négresko - Paulet', 'Promenade Pompidou Palm Beach', 'Borne Test Nantes 1', 'Flammariion Montrichet', 'Cantini Rouet', 'Arenç Chanterac', 'Canebière Dugommier', 'Canebière Saint Ferreol', 'Sakakini - Saint Pierre', 'Blancarde', 'Gambetta - Lafayette', 'Corse Saint Maurice', 'Pierre Puget Breteuil', 'Hotel De Ville', 'Plaine - Bibliotheque', 'Stalingrad - Réformés', 'It Plaisir', 'Roosevelt - Oran', '153 Paradis', 'Catalans Corniche', and 'Test It'.

The 'Applied Steps' pane on the right shows the following steps:

- replaced values (case-no ...
- Replaced Value3(true -yes ...
- Inserted Capitalize Each W...
- Inserted Coll(Capitalize e...
- Removed Columns2
- Renamed Columns(Station...
- Inserted Capitalize Each W...
- Reordered Columns
- Removed Columns3
- Renamed Columns-city na...
- Reordered Columns1
- Renamed Columns
- Split Column by Delimiter(...
- Changed Type6
- Renamed Columns1(date,t...
- Filtered Rows1
- Removed Duplicates
- Replaced Value
- Replaced Value1

- Build a logical **data model**.
 - Set up a fact table.
 - Create dimension tables and

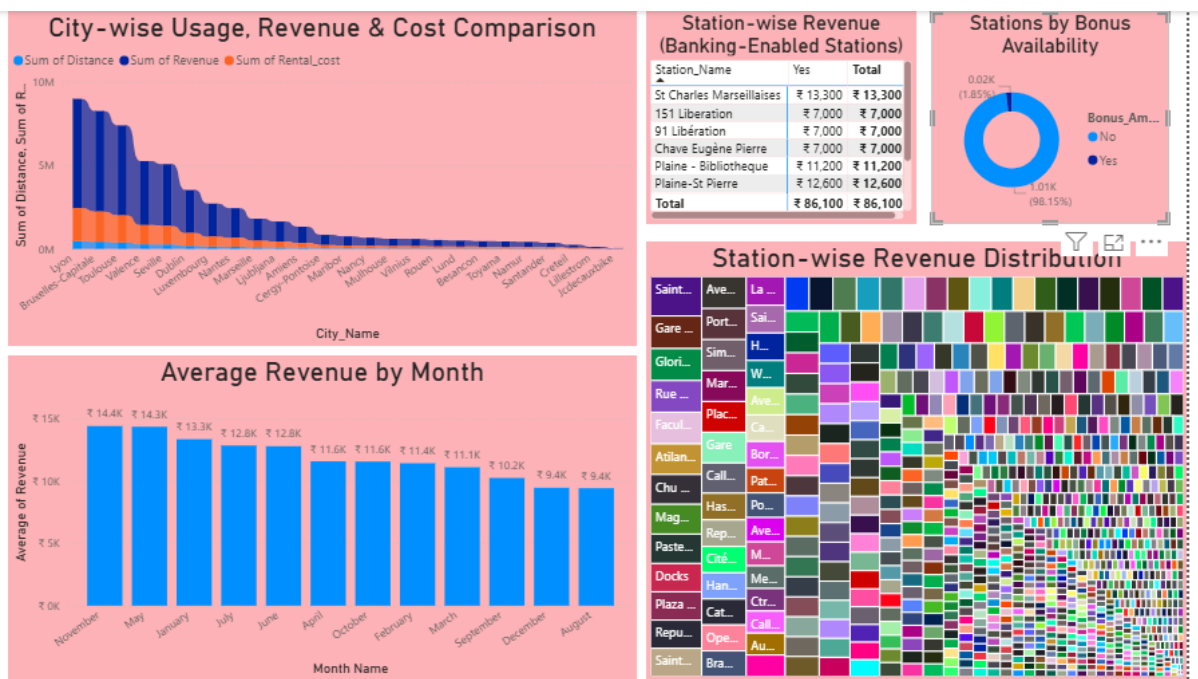
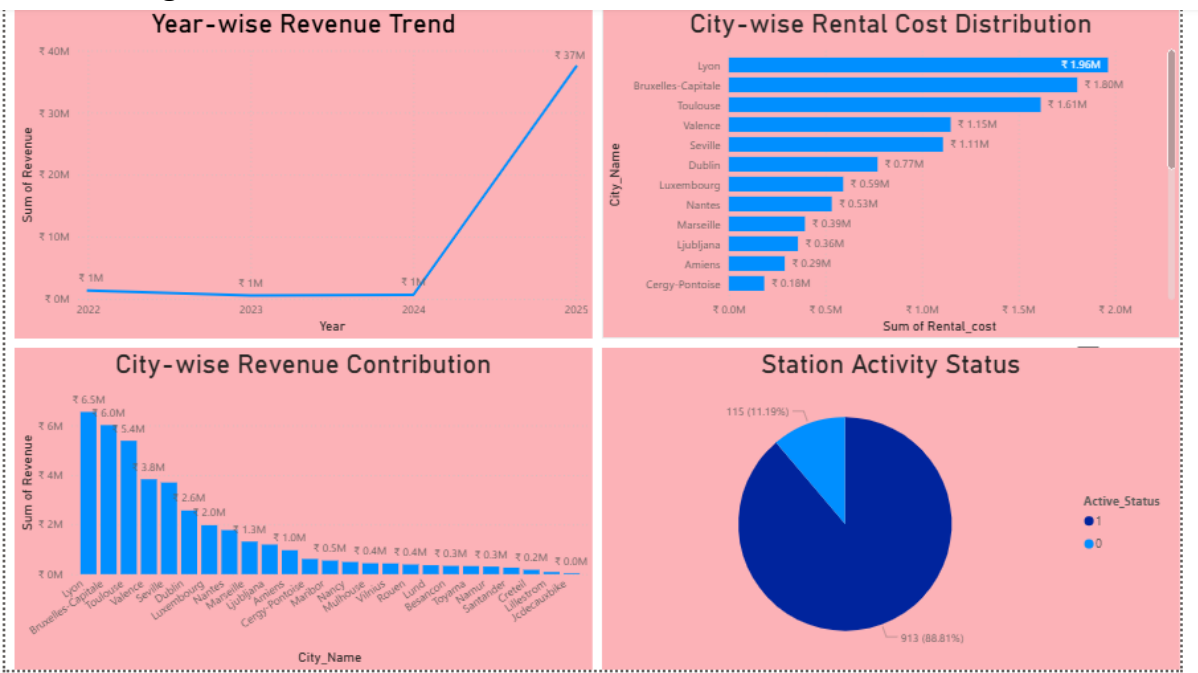
Defining the relationships between the fact and dimension tables



- Create **DAX measures** and calculated columns (wherever needed).



● Design interactive visuals and dashboard



Consolidated dashboard

